

ROMIT KUMAR PAL

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Software Development Engineer with 1+ year of production experience building real-time voice AI systems, LLM-powered applications, and scalable REST APIs. Proven expertise delivering end-to-end conversational AI pipelines with sub-500ms latency using Python, FastAPI, WebRTC, Deepgram, LangChain, and Docker. Experienced across full-stack development in Python, .NET Core, Flutter, and React. 2× Patent holder with published research in Computer Vision and NLP.

EXPERIENCE

Leap91 Technologies Pvt. Ltd.

Jan 2026 – Present

Software Development Engineer — Full Stack AI Engineer

Pune, Maharashtra (Remote)

- Developed core backend features for **Crewgle AI**, a generative AI recruitment platform, designing modular REST API endpoints and microservices using Python and FastAPI; improved developer throughput and system reliability through scalable, testable architecture.
- Engineered a production-grade real-time conversational AI voice pipeline using Pipecat, integrating WebRTC (Daily.co), Deepgram STT, OpenAI GPT-4, ElevenLabs TTS, and Silero VAD — achieving sub-500ms end-to-end voice interaction latency with natural turn-taking and robust error recovery.
- Designed and deployed a RAG-powered voice assistant for GBS Organization using LangChain and FAISS vector database, enabling context-aware, multi-turn AI conversations grounded in proprietary knowledge bases with accurate document retrieval.
- Developed a fully local Speech-to-Speech inference stack (Ollama + Whisper + Kokoro-82M TTS) eliminating third-party API costs and enabling offline, privacy-sensitive enterprise deployments.
- Integrated WebRTC (Daily.co) real-time video infrastructure for concurrent AI-human interview sessions with session lifecycle management, error recovery, and horizontal scalability via Docker containerization.

SigiloTech

Jul 2025 – Dec 2025

Software Engineer Intern — Backend Developer

Kolkata, West Bengal (On-site)

- Designed and scaled 50+ controller-based RESTful APIs using .NET Core (C#) for the IMPKT Project, supporting critical business workflows including authentication, data management, and third-party integrations end-to-end.
- Architected SQL database schema and high-performance data access layer using Dapper Micro-ORM with parameterized queries, optimizing throughput for concurrent, high-volume database operations.
- Built an intelligent file management system integrating Azure Blob Storage with GPT-4 Vision for automatic description generation, reducing manual content-tagging effort to zero and cutting review time by ~70%.
- Implemented .NET Core middleware pipeline for structured request logging, centralized exception handling, and JWT-based stateless authentication and role-based authorization across all service endpoints.

PROJECTS

GitaVerse: Voice of KrishnaAI [Live on Google Play] [App Link](#) | Flutter • FastAPI • Pipecat • WebRTC • Deepgram • OpenAI • ElevenLabs

- Led end-to-end architecture of a production cross-platform voice AI mobile app (Flutter + FastAPI + WebRTC), live on Google Play — enabling real-time spiritual guidance conversations with a Krishna-inspired AI persona at scale.
- Designed low-latency speech-to-speech pipeline: Silero VAD → Deepgram STT → OpenAI LLM → ElevenLabs TTS targeting ~1–1.5s turn response; implemented WebRTC session handling, active session caps, and graceful fallback mechanisms.
- Owned full backend API design (WebRTC signaling, content modules, chapter wisdom, meditation flows) and led RevenueCat monetization with server-side quota and credit tracking (free trial + paid top-ups).
- Shipped production Docker infrastructure: health checks, CORS, rate limiting, Firebase Analytics, Google Ads, multi-language support (English/Hindi/Hinglish), and daily TTS audio caching.

Fashion-Saathi — AI Wedding Organizer [App Link](#) | React • Node.js • Express • MongoDB • NLP

- Designed an NLP-driven AI recommendation engine for personalized wedding styling (attire, makeover, hairstyle), processing multi-attribute user profiles through a natural-language query interface on React, Node.js, and MongoDB.

TECHNICAL SKILLS

Languages: Python, Dart (Flutter), Java

Frameworks: FastAPI, .NET Core, LangChain, Pipecat, React.js, Flutter

AI / GenAI: LLM Agents, RAG Pipelines, OpenAI API, Deepgram STT, Whisper, ElevenLabs TTS, Ollama, HuggingFace

Databases: FAISS, Pinecone, PostgreSQL, MongoDB, MSSQL, Vector DBs

Real-Time & APIs: WebRTC (Daily.co), WebSockets, REST APIs, JWT, Socket.io, RevenueCat

RESEARCH, PATENTS & ACHIEVEMENTS

- Patent Co-Applicant: “Kanoon Mitra” — AI/NLP legal chatbot for intelligent legal query interpretation using knowledge graphs (Application No: 202331049760A).
- Patent Co-Applicant: “Chakravayuh” — AI-powered real-time video surveillance detecting theft and antisocial behaviour using Computer Vision (Application No: 202331049753A).
- Published Research: “An AI Based Integrated Framework for Motion Activated Facial Recognition” — ICIDA, International Springer Conference on Innovations in Data Analytics. [Paper Link](#)
- Hackathon Finalist: CODESPIRE 2023 — Ranked 7th All-India in a national-level competitive coding and product innovation hackathon. [Certificate](#)

EDUCATION

Brainware University, Kolkata

Aug 2021 – Jun 2025

B.Tech, Computer Science and Engineering (AI-ML Specialization) • CGPA: 9.50 / 10.00